

Figure 2.19 Schematic illustration of metallic bonding.

per atom. Figure 2.19 illustrates metallic bonding. The free electrons shield the positively charged ion cores from the mutually repulsive electrostatic forces that they would otherwise exert upon one another; consequently, the metallic bond is nondirectional in character. In addition, these free electrons act as a "glue" to hold the ion cores together. Bonding energies and melting temperatures for several metals are listed in Table 2.3. Bonding may be weak or strong; energies range from 62 kJ/mol for mercury to 850 kJ/mol for tungsten. Their respective melting temperatures are −39°C and 3414°C (−39°F and 6177°F).

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What Is Metallic Bonding?

Metallic bonding is found in the periodic table for Group IA and IIA elements and, in fact, for all elemental metals.

Metals are good conductors of both electricity and heat as a consequence of their free electrons (see Sections 18.5, 18.6, and 19.4). Furthermore, in Section 7.4, we note that at room temperature, most metals and their alloys fail in a ductile manner—that is, fracture occurs after the materials have experienced significant degrees of permanent deformation. This behavior is explained in terms of a deformation mechanism (Section 7.2), which is implicitly related to the characteristics of the metallic bond.

Concept Check 2.3 Explain why covalently bonded materials are generally less dense than ionically or metallically bonded ones.

[*The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister (Student Companion Site).*]

2.7 SECONDARY BONDING OR VAN DER WAALS BONDING

secondary bond
van der Waals bond

Secondary bonds, or **van der Waals** (physical) **bonds,** are weak in comparison to the primary or chemical bonds; bonding energies range between about 4 and 30 kJ/mol. Secondary bonding exists between virtually all atoms or molecules, but its presence may be obscured if any of the three primary bonding types is present. Secondary bonding is evidenced for the inert gases, which have stable electron structures. In addition, secondary (or *inter*molecular) bonds are possible between atoms or groups of atoms, which themselves are joined together by primary (or *intra*molecular) ionic or covalent bonds.

dipole

Secondary bonding forces arise from atomic or molecular **dipoles.** In essence, an electric dipole exists whenever there is some separation of positive and negative portions of an atom or molecule. The bonding results from the coulombic attraction between the positive end of one dipole and the negative region of an adjacent one, as indicated in Figure 2.20. Dipole interactions occur between induced dipoles, between induced dipoles and polar molecules (which have permanent dipoles), and between polar molecules. **Hydrogen bonding,** a special type of secondary bonding, is found to exist

hydrogen bonding

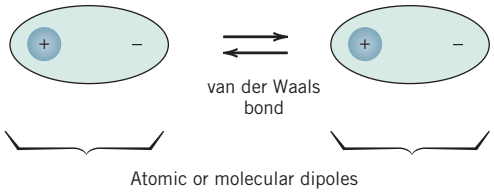


Figure 2.20 Schematic illustration of van der Waals bonding between two dipoles.

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What Is a Dipole?

between some molecules that have hydrogen as one of the constituents. These bonding mechanisms are discussed briefly next.

Fluctuating Induced Dipole Bonds

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What Is van der Waals Bonding?

A dipole may be created or induced in an atom or molecule that is normally electrically symmetric—that is, the overall spatial distribution of the electrons is symmetric with respect to the positively charged nucleus, as shown in Figure 2.21*a*. All atoms experience constant vibrational motion that can cause instantaneous and short-lived distortions of this electrical symmetry for some of the atoms or molecules and the creation of small electric dipoles. One of these dipoles can in turn produce a displacement of the electron distribution of an adjacent molecule or atom, which induces the second one also to become a dipole that is then weakly attracted or bonded to the first (Figure 2.21*b*); this is one type of van der Waals bonding. These attractive forces, which are temporary and fluctuate with time, may exist between large numbers of atoms or molecules.

The liquefaction and, in some cases, the solidification of the inert gases and other electrically neutral and symmetric molecules such as H_2 and Cl_2 are realized because of this type of bonding. Melting and boiling temperatures are extremely low in materials for which induced dipole bonding predominates; of all possible intermolecular bonds, these are the weakest. Bonding energies and melting temperatures for argon, krypton, methane, and chlorine are also tabulated in Table 2.3.

Polar Molecule–Induced Dipole Bonds

polar molecule

Permanent dipole moments exist in some molecules by virtue of an asymmetrical arrangement of positively and negatively charged regions; such molecules are termed **polar molecules.** Figure 2.22*a* shows a schematic representation of a hydrogen chloride molecule; a permanent dipole moment arises from net positive and negative charges that are respectively associated with the hydrogen and chlorine ends of the HCl molecule.

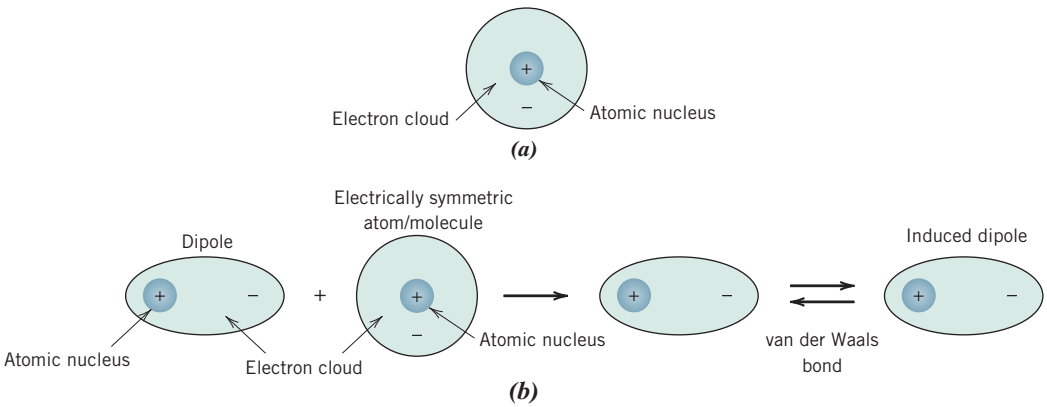


Figure 2.21 Schematic representations of (*a*) an electrically symmetric atom and (*b*) how an electric dipole induces an electrically symmetric atom/molecule to become a dipole—also the van der Waals bond between the dipoles.

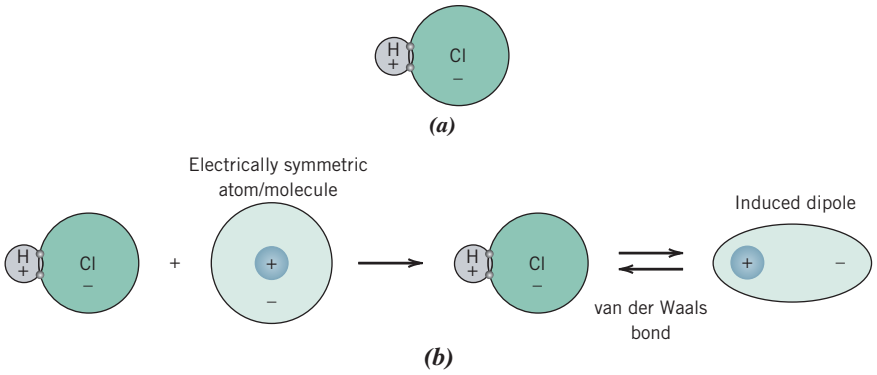


Figure 2.22 Schematic representations of (*a*) a hydrogen chloride molecule (dipole) and (*b*) how an HCl molecule induces an electrically symmetric atom/molecule to become a dipole—also the van der Waals bond between these dipoles.

Polar molecules can also induce dipoles in adjacent nonpolar molecules, and a bond forms as a result of attractive forces between the two molecules; this bonding scheme is represented schematically in Figure 2.22*b*. Furthermore, the magnitude of this bond is greater than for fluctuating induced dipoles.

Permanent Dipole Bonds

Coulombic forces also exist between adjacent polar molecules as in Figure 2.20. The associated bonding energies are significantly greater than for bonds involving induced dipoles.

The strongest secondary bonding type, the hydrogen bond, is a special case of polar molecule bonding. It occurs between molecules in which hydrogen is covalently bonded to fluorine (as in HF), oxygen (as in H_2O), or nitrogen (as in NH_3). For each H—F, H—O, or H—N bond, the single hydrogen electron is shared with the other atom. Thus, the hydrogen end of the bond is essentially a positively charged bare proton unscreened by any electrons. This highly positively charged end of the molecule is capable of a strong attractive force with the negative end of an adjacent molecule, as demonstrated in Figure 2.23 for HF. In essence, this single proton forms a bridge between two negatively charged atoms. The magnitude of the hydrogen bond is generally greater than that of the other types of secondary bonds and may be as high as 51 kJ/mol, as shown in Table 2.3. Melting and boiling temperatures for hydrogen fluoride, ammonia, and water are abnormally high in light of their low molecular weights, as a consequence of hydrogen bonding.

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What Are the Differences between Ionic, Covalent, Metallic, and van der Waals Types of Bonding?

In spite of the small energies associated with secondary bonds, they nevertheless are involved in a number of natural phenomena and many products that we use on a daily basis. Examples of physical phenomena include the solubility of one substance in another, surface tension and capillary action, vapor pressure, volatility, and viscosity. Common applications that make use of these phenomena include *adhesives*—van der Waals bonds form between two surfaces so that they adhere to one another (as discussed in the chapter opener for this chapter); *surfactants*—compounds that lower the surface tension of a liquid and are found in soaps, detergents, and foaming agents; *emulsifiers*—substances that, when added to two immiscible materials (usually liquids), allow particles of one material to be suspended in another (common emulsions include sunscreens, salad dressings, milk, and mayonnaise); and *desiccants*—materials that form hydrogen bonds with water molecules (and remove moisture from closed containers—e.g., small packets that are often found in cartons of packaged goods); and finally, the strengths, stiffnesses, and softening temperatures of polymers, to some degree, depend on secondary bonds that form between chain molecules.

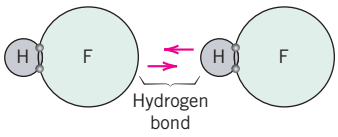


Figure 2.23 Schematic representation of hydrogen bonding in hydrogen fluoride (HF).

MATERIAL OF IMPORTANCE 2.1

Water (Its Volume Expansion Upon Freezing)

Upon freezing (i.e., transforming from a liquid to a solid upon cooling), most substances experience an increase in density (or, correspondingly, a decrease in volume). One exception is water, which exhibits the anomalous and familiar expansion upon freezing—approximately 9 volume percent expansion. This behavior may be explained on the basis of hydrogen bonding. Each H_2O molecule has two hydrogen atoms that can bond to oxygen atoms; in addition, its single O atom can bond to two hydrogen atoms of other H_2O molecules. Thus, for solid ice, each water molecule participates in four hydrogen bonds, as shown in the three-dimensional schematic of Figure 2.24*a;* here, hydrogen bonds are denoted by dashed lines, and each water molecule has 4 nearest-neighbor molecules. This is a relatively open structure—that is, the molecules are not closely packed together—and as a result, the density is comparatively low. Upon melting, this structure is partially destroyed, such that the water molecules become more closely packed together (Figure 2.24*b*)—at room temperature, the average number of nearest-neighbor water molecules has increased to approximately 4.5; this leads to an increase in density.

Consequences of this anomalous freezing phenomenon are familiar; it explains why icebergs float; why, in cold climates, it is necessary to add antifreeze to an automobile's cooling system (to keep the engine block from cracking); and why freeze–thaw cycles break up the pavement in streets and cause potholes to form.



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A watering can that ruptured along a side panel—bottom panel seam. Water that was left in the can during a cold late-autumn night expanded as it froze and caused the rupture.

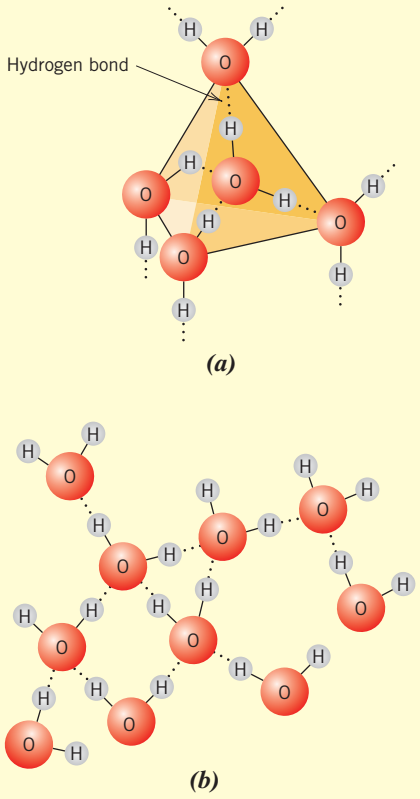


Figure 2.24 The arrangement of water (H_2O) molecules in (*a*) solid ice and (*b*) liquid water.

2.8 MIXED BONDING

Sometimes it is illustrative to represent the four bonding types—ionic, covalent, metallic, and van der Waals—on what is called a *bonding tetrahedron*—a three-dimensional tetrahedron with one of these "extreme" types located at each vertex, as shown in Figure 2.25*a*. Furthermore, we should point out that for many real materials, the atomic bonds are mixtures of two or more of these extremes (i.e., *mixed bonds*). Three mixed-bond types—covalent–ionic, covalent–metallic, and metallic–ionic—are also included on edges of this tetrahedron; we now discuss each of them.

For mixed covalent–ionic bonds, there is some ionic character to most covalent bonds and some covalent character to ionic ones. As such, there is a continuum between these two extreme bond types. In Figure 2.25*a*, this type of bond is represented between the ionic and covalent bonding vertices. The degree of either bond type depends on the relative positions of the constituent atoms in the periodic table (see Figure 2.8) or the difference in their electronegativities (see Figure 2.9). The wider the separation (both horizontally—relative to Group IVA—and vertically) from the lower left to the upper right corner (i.e., the greater the difference in electronegativity), the more ionic is the bond. Conversely, the closer the atoms are together (i.e., the smaller the difference in electronegativity), the greater is the degree of covalency. Percent ionic character (%IC) of a bond between elements A and B (A being the most electronegative) may be approximated by the expression

$$\%\text{IC} = \{1 - \exp[-(0.25)(X_A - X_B)^2]\} \times 100 \tag{2.16}$$

where X_A and X_B are the electronegativities for the respective elements.

Another type of mixed bond is found for some elements in Groups IIIA, IVA, and VA of the periodic table (viz., B, Si, Ge, As, Sb, Te, Po, and At). Interatomic bonds for these elements are mixtures of metallic and covalent, as noted on Figure 2.25*a*. These materials are called the *metalloids* or *semi-metals*, and their properties are intermediate between the metals and nonmetals. In addition, for Group IV elements, there is a gradual transition from covalent to metallic bonding as one moves vertically down this column—for example, bonding in carbon (diamond) is purely covalent, whereas for tin and lead, bonding is predominantly metallic.

Mixed metallic–ionic bonds are observed for compounds composed of two metals when there is a significant difference between their electronegativities. This means that

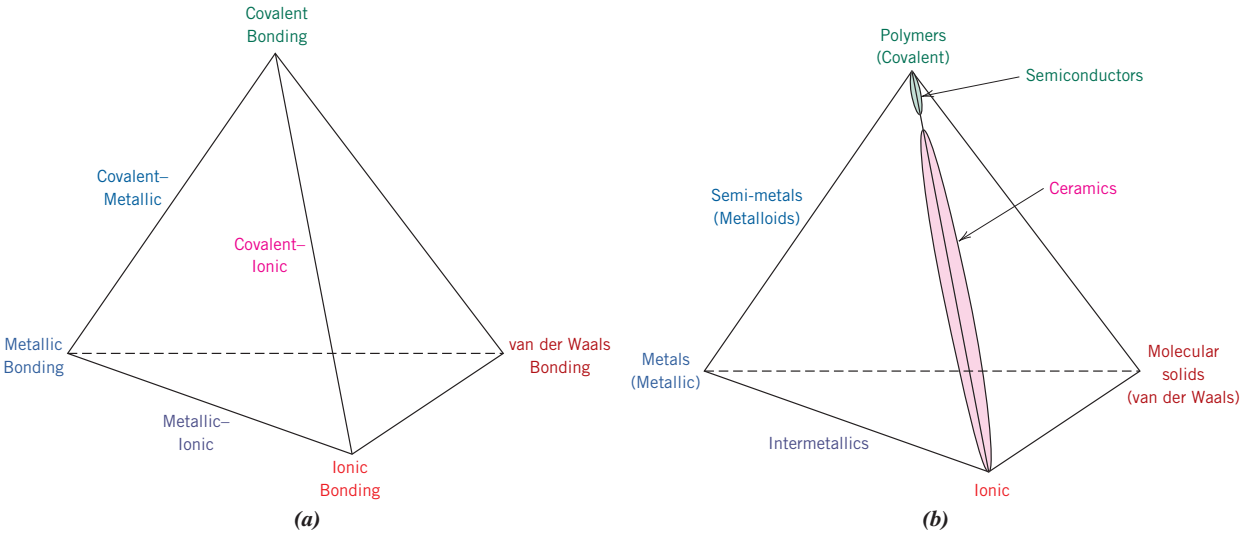


Figure 2.25 (*a*) Bonding tetrahedron: Each of the four extreme (or pure) bonding types is located at one corner of the tetrahedron; three mixed bonding types are included along tetrahedron edges. (*b*) Material-type tetrahedron: correlation of each material classification (metals, ceramics, polymers, etc.) with its type(s) of bonding.